

RE-009: Understanding Nutritional and Ingredient-Based Implications of Dietary Restrictions

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1. Impact of dietary restrictions on Macronutrients and Micronutrients:

Micronutrients and macronutrients are influenced by restrictions in diet.

- Vegan/Vegetarian diets: Vegan or Vegetarian diets often have less protein intake and many nutrients like vitamin B12, iron and omega-3 fatty acids are absent.
- Gluten-free diet: This diet has less amount of fibre and vitamin-B because of whole grains are not included in this diet.
- Lactose-free diet: Dairy products are absent in this diet, so it led to reduced intake of calcium and vitamin D.

Nutrient Impact Overview:

Diet Type	Reduced Nutrients
Vegan	Vitamin B12, Iron, Omega-3, Zinc
Vegetarian	Vitamin B12 (partial), Iron
Gluten-free	Fibre, B Vitamins, Iron
Lactose-free	Calcium, Vitamin D, Riboflavin



Figure 1: Comparison of Nutrient Impact Across Dietary Restrictions

2. Ingredients that are commonly restricted across these diets

Diet Type	Restricted Ingredients
Vegan	Meat, dairy, eggs, honey, Milk
Vegetarian	Meat, fish
Gluten-free	Wheat, barley, rye
Lactose-free	Milk, cheese, butter

3. Potential Nutritional Risks and Deficiencies:

These restrictions can lead to deficiencies of several nutrients if not included through alternatives.

- Vitamin B12 deficiency (very common in vegan diets)
- Iron deficiency (vegetarian or vegan diets)
- Calcium deficiency (lactose-free diets)
- Fibre deficiency (gluten-free diets if not balanced properly)

These deficiencies can affect the energy levels, bone health and overall health of a human being.

These deficiencies can affect:

- Production of energy
- Immune function
- Bone density
- Cognitive performance

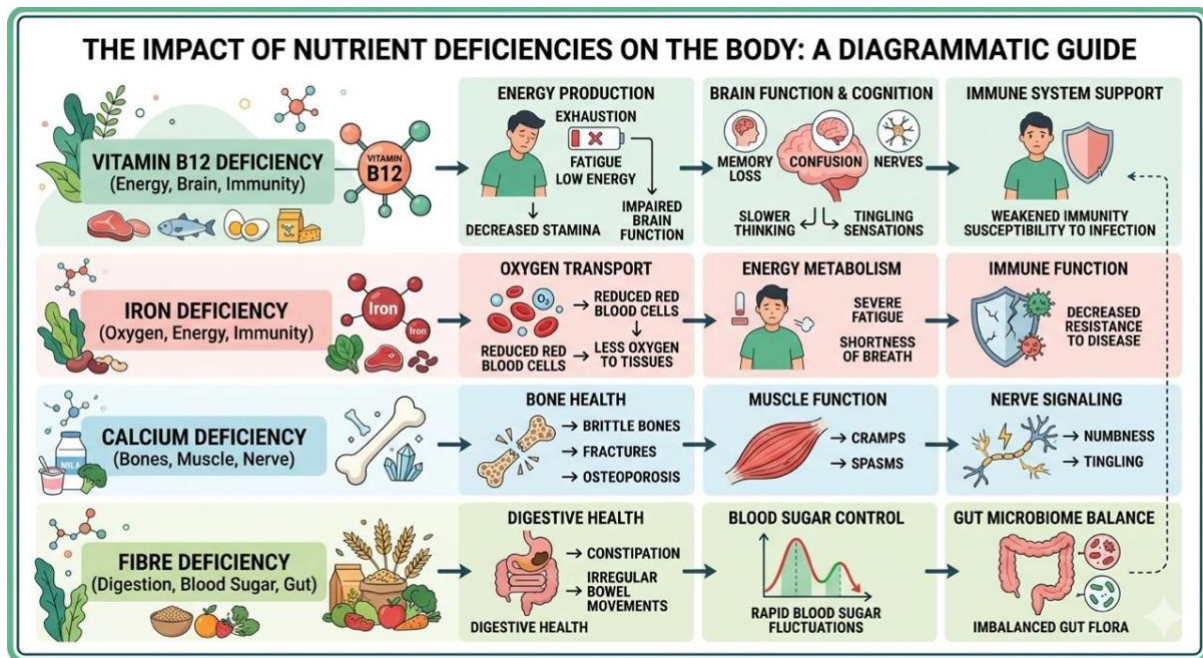


Figure 3: Effects of Nutritional Deficiencies on the Human Body

4. Adaptation Strategies to meet dietary requirements:

People can use some strategies to maintain their overall nutritional balance.

- Substitutions:

Strategy Example	Suitable Diets
Plant-based milk (soy, almond, oat)	Vegan, Lactose-free
Gluten-free grains (rice, quinoa)	Gluten-free
Legumes replacing meat protein	Vegan, Vegetarian
Nut-based yoghurts	Vegan, Lactose-free

- Fortified Foods:

Fortified Product	Nutrient Provided	Suitable Diets
Plant-based milk	Calcium, Vitamin D	Vegan, Lactose-free
Breakfast cereals	Vitamin B12, Iron	Vegan, Vegetarian
Gluten-free bread	Fibre, B vitamins	Gluten-free

- Supplements:

- Diet

Supplement	Used For Deficiency	Common Diets
Vitamin B12	Prevents neurological issues	Vegan
Iron	Treats anaemia	Vegan, Vegetarian
Vitamin D	Bone health	Lactose-free, Vegan
Omega-3	Heart and brain health	Vegan

Planning:

- Combining foods, for example, legumes + grains can balance protein requirements in the body.

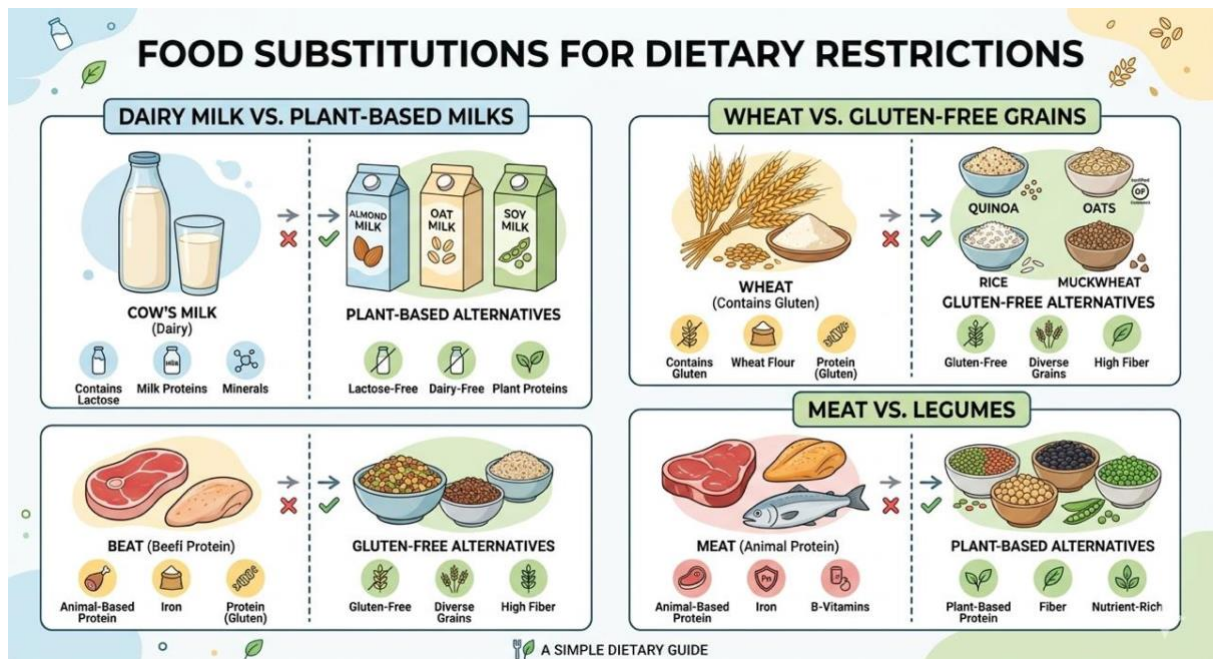


Figure 2: Common Food Substitutions for Dietary Restrictions

5. Conclusion

Dietary restriction can affect one's health by limiting the nutrition and certain ingredients. While these restrictions can cause deficiencies of few nutrients, but it can be prevented by adapting substitutions, fortified foods, supplements and proper planning. These implications are important to understand for designing the functioning of systems like Food Remedy API that can help to give proper dietary recommendation.

6. References:

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